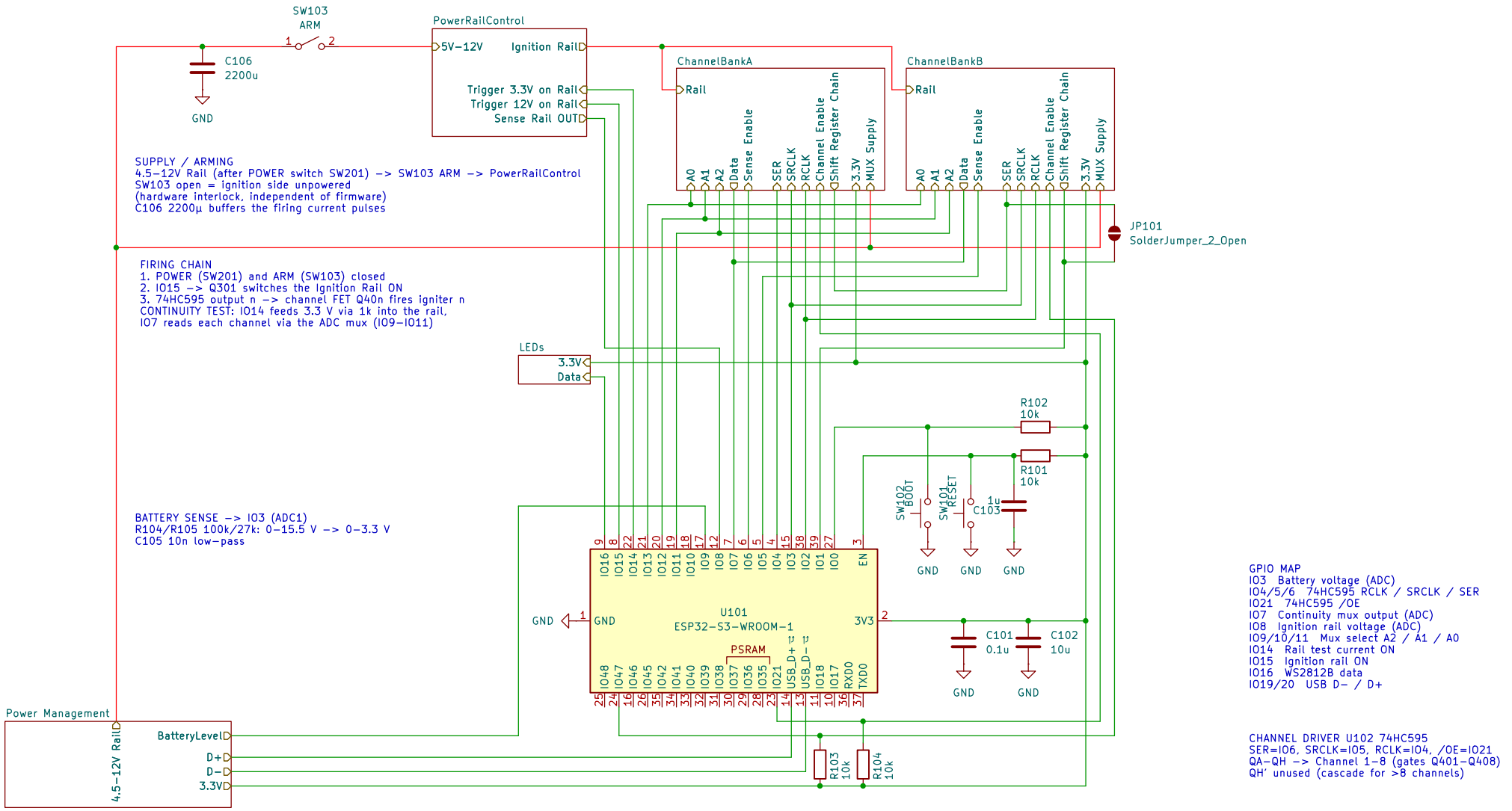
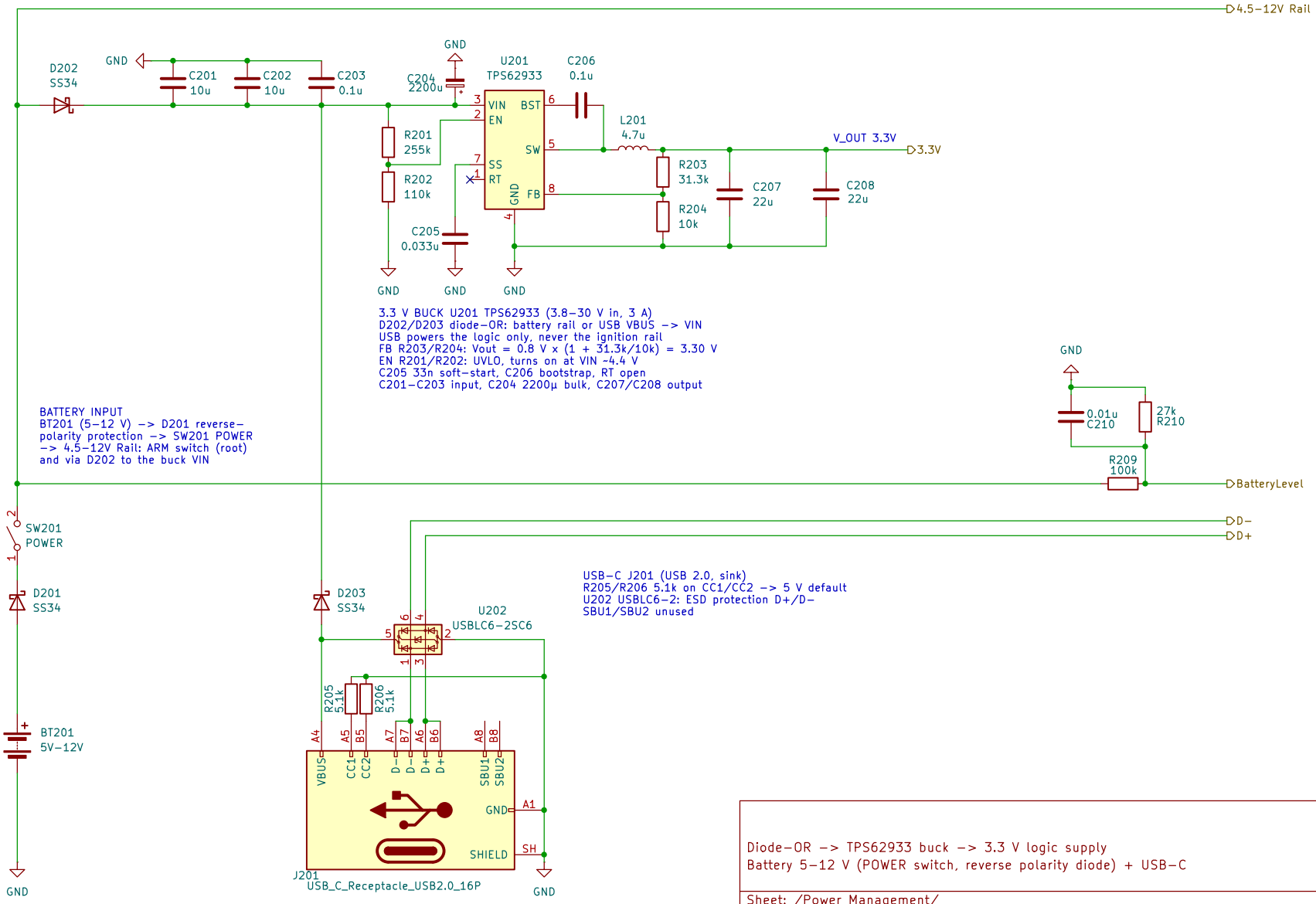
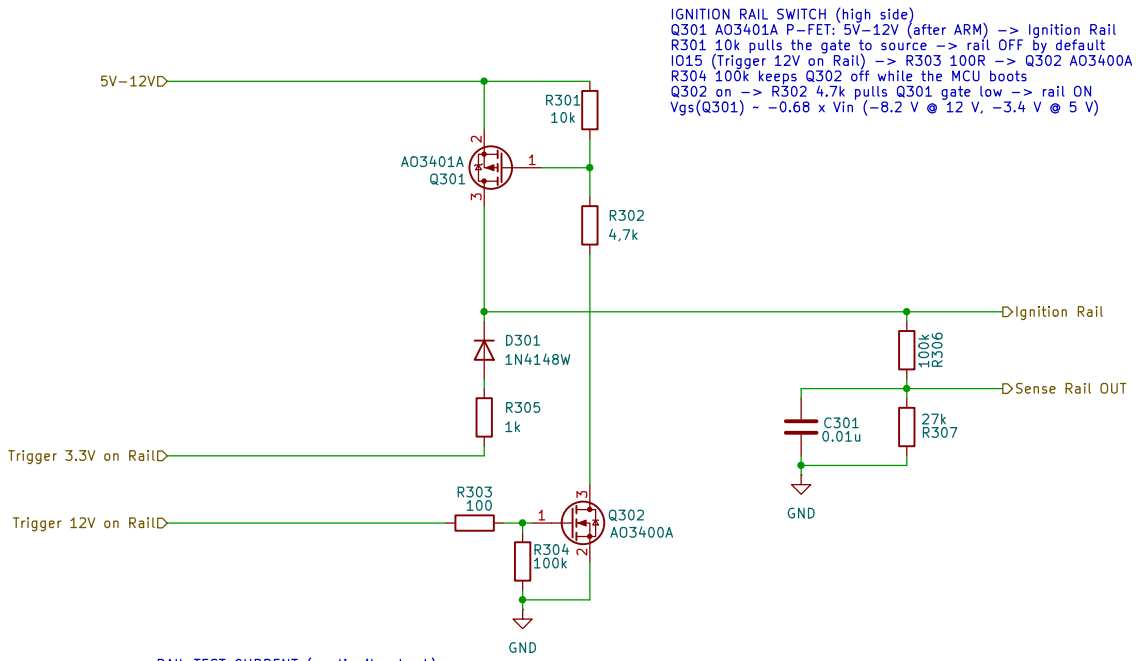

SW103
ARM
C106
2200u
GND
PowerRailControl
5V–12V
Ignition Rail
Trigger 3.3V on Rail
Trigger 12V on Rail
Sense Rail OUT
ChannelBankA
ChannelBankB
Rail
A0
A1
A2
Data
Sense Enable
SER
SRCLK
RCLK
Channel Enable
Shift Register Chain
3.3V
MUX Supply
JP101
SolderJumper_2_Open
SUPPLY / ARMING
4.5–12V Rail (after POWER switch SW201) –> SW103 ARM –> PowerRailControl
SW103 open = ignition side unpowered
(hardware interlock, independent of firmware)
C106 2200µ buffers the firing current pulses
FIRING CHAIN
1. POWER (SW201) and ARM (SW103) closed
2. IO15 –> Q301 switches the Ignition Rail ON
3. 74HC595 output n –> channel FET Q40n fires igniter n
CONTINUITY TEST: IO14 feeds 3.3 V via 1k into the rail,
IO7 reads each channel via the ADC mux (IO9–IO11)
LEDs
3.3V
Data
R102
10k
R101
10k
SW102
BOOT
SW101
RESET
C103
1u
GND
BATTERY SENSE –> IO3 (ADC1)
R104/R105 100k/27k: 0–15.5 V –> 0–3.3 V
C105 10n low–pass
U101
ESP32–S3–WROOM–1
PSRAM
3V3
EN
C101
0.1u
C102
10u
Power Management
4.5–12V Rail
BatteryLevel
D+
D–
3.3V
R103
10k
R104
10k
GPIO MAP
IO3 Battery voltage (ADC)
IO4/5/6 74HC595 RCLK / SRCLK / SER
IO21 74HC595 /OE
IO7 Continuity mux output (ADC)
IO8 Ignition rail voltage (ADC)
IO9/10/11 Mux select A2 / A1 / A0
IO14 Rail test current ON
IO15 Ignition rail ON
IO16 WS2812B data
IO19/20 USB D– / D+
CHANNEL DRIVER U102 74HC595
SER=IO6, SRCLK=IO5, RCLK=IO4, /OE=IO21
QA–QH –> Channel 1–8 (gates Q401–Q408)
QH' unused (cascade for >8 channels)
MCU U101 ESP32–S3–WROOM–1
SW101 RESET: EN with RC R101/C103 (~10 ms power–on delay)
SW102 BOOT: IO0 low at reset –> download mode, R102 pull–up
C101 10µ / C102 100n: 3V3 bulk + decoupling at the module
USB D+/D– from J201 (ESD U202) –> native USB (flash + CDC)
Refs: <page><nn> 1xx root, 2xx power, 3xx rail ctrl, 4xx channels, 5xx LEDs, 6xx continuity
ESP32–S3 node: MCU, 74HC595 channel driver, battery sense, ARM switch
Sheet: /
File: MK4–Proto.kicad_sch
Title: MK4–Proto (Zündanlage MK4 Prototype)
Size: A4
Date: 2026–10–02
Rev: 0
KiCad E.D.A. 10.0.6
Id: 1/10


4.5–12V Rail
D202
SS34
GND
C201
10u
C202
10u
C203
0.1u
C204
2200u
U201
TPS62933
C206
0.1u
VIN
EN
SS
RT
GND
BST
SW
FB
L201
4.7u
V_OUT 3.3V
3.3V
R201
255k
R202
110k
R203
31.3k
R204
10k
C205
0.033u
C207
22u
C208
22u
3.3 V BUCK U201 TPS62933 (3.8–30 V in, 3 A)
D202/D203 diode-OR: battery rail or USB VBUS -> VIN
USB powers the logic only, never the ignition rail
FB R203/R204: Vout = 0.8 V x (1 + 31.3k/10k) = 3.30 V
EN R201/R202: UVLO, turns on at VIN ~4.4 V
C205 33n soft-start, C206 bootstrap, RT open
C201–C203 input, C204 2200µ bulk, C207/C208 output
BATTERY INPUT
BT201 (5–12 V) -> D201 reverse-polarity protection -> SW201 POWER -> 4.5–12V Rail: ARM switch (root) and via D202 to the buck VIN
SW201
POWER
D201
SS34
BT201
5V–12V
D203
SS34
U202
USBLC6-2SC6
R205
5.1k
R206
5.1k
USB-C J201 (USB 2.0, sink)
R205/R206 5.1k on CC1/CC2 -> 5 V default
U202 USBLC6-2: ESD protection D+/D-
SBU1/SBU2 unused
D-
D+
VBUS
CC1
CC2
SBU1
SBU2
SHIELD
SH
A1
J201
USB_C_Receptacle_USB2.0_16P
C210
0.01u
R210
27k
R209
100k
BatteryLevel
Diode-OR -> TPS62933 buck -> 3.3 V logic supply
Battery 5–12 V (POWER switch, reverse polarity diode) + USB-C
Sheet: /Power Management/
File: PowerManagement.kicad_sch
Title: Power Management
Size: A4
Date: 2026-10-30
Rev:
KiCad E.D.A. 10.0.6
Id: 2/10

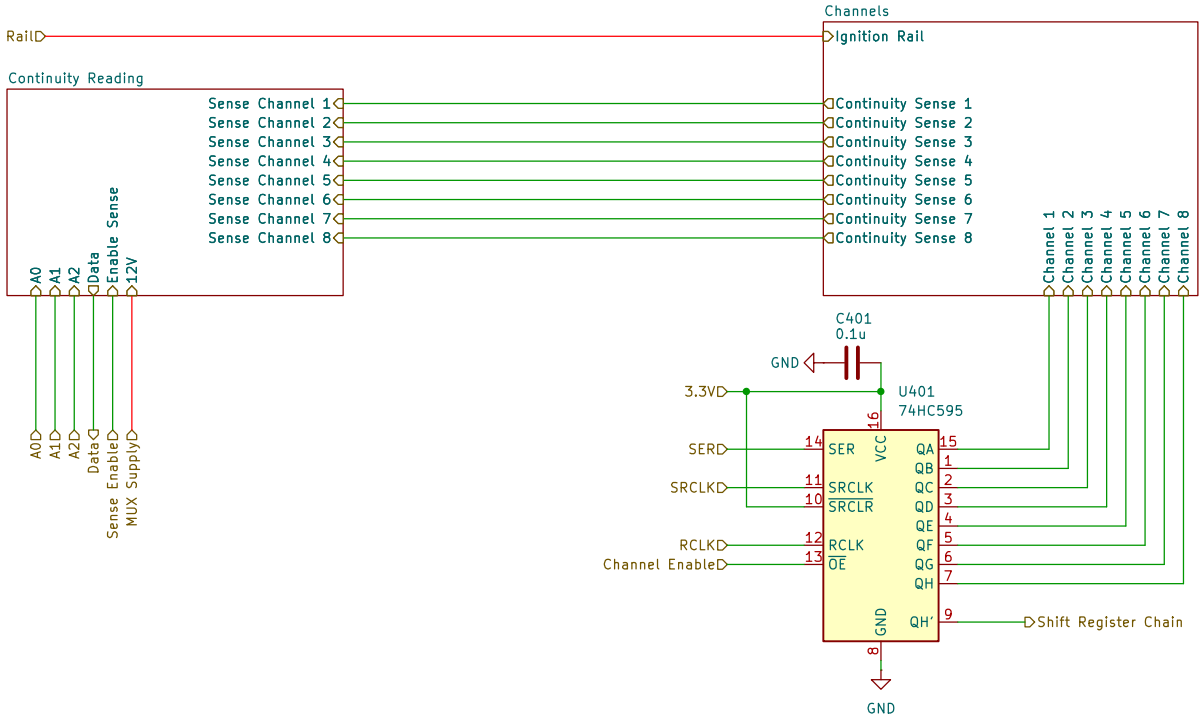


Current-limited 3.3 V test feed for continuity check (IO14)
High-side P-FET switch for the ignition rail (IO15)

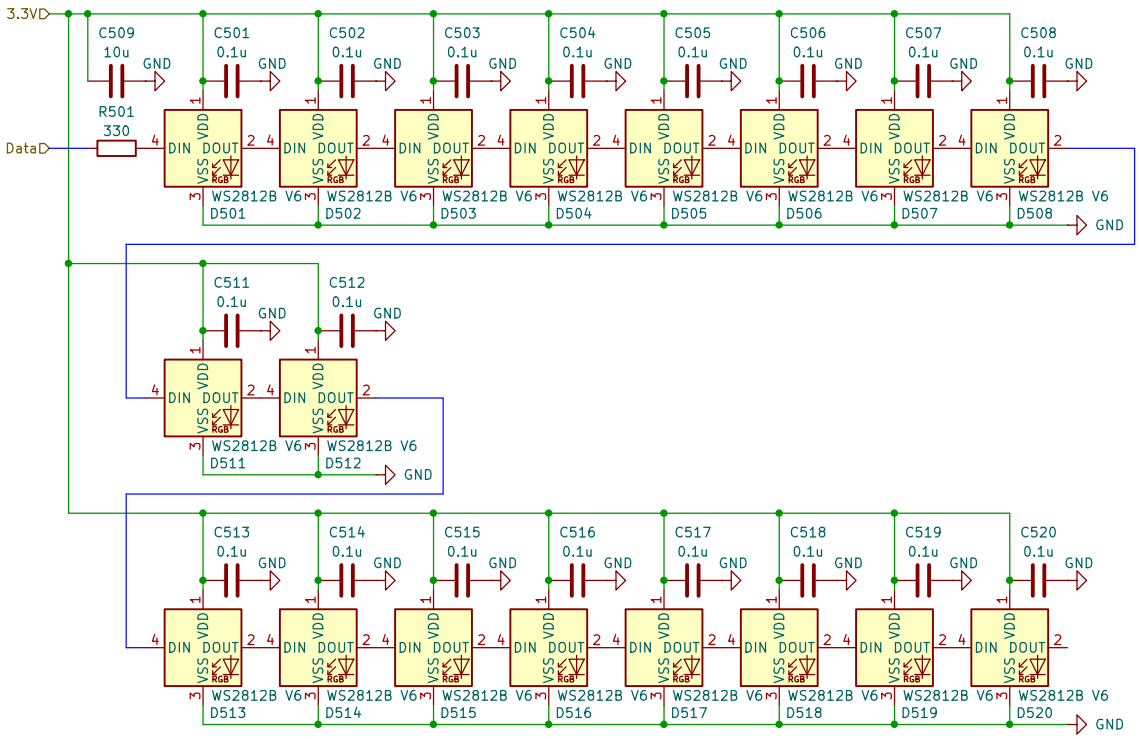
Sheet: /PowerRailControl/
File: testing.kicad_sch

Title: Ignition & Test Rail Control

Size: A4	Date: 2026-10-24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/10



Sheet: /ChannelBankA/		
File: channelBank.kicad_sch		
Title: Channel Bank		
Size: A4	Date: 2026-11-01	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/10



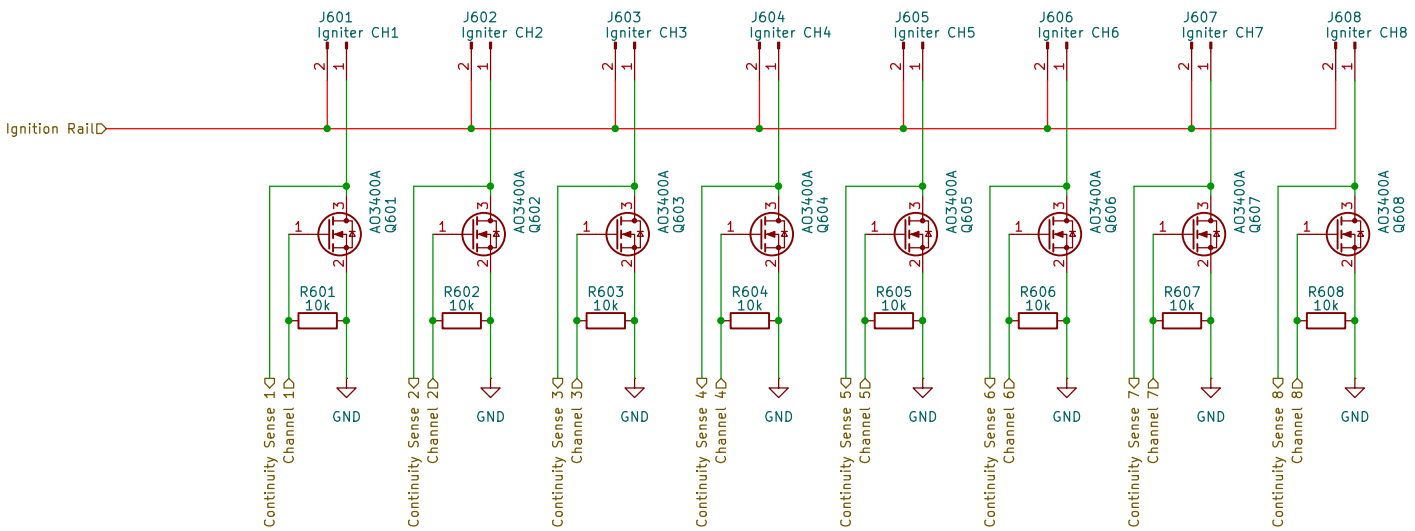
STATUS / CHANNEL LEDs
IO16 -> R501 330R series resistor -> WS2812B chain D501 ... D510
Chain order = LED index in firmware (D501 = index 0, D510 = index 9)
C501–C510: 100n decoupling, one per LED

WS2812B chain, data from IO16
2 Status Leds
8 Channel Leds

Sheet: /LEDs/
File: leds.kicad_sch

Title: LEDs

Size: A4	Date: 2026-10-24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/10



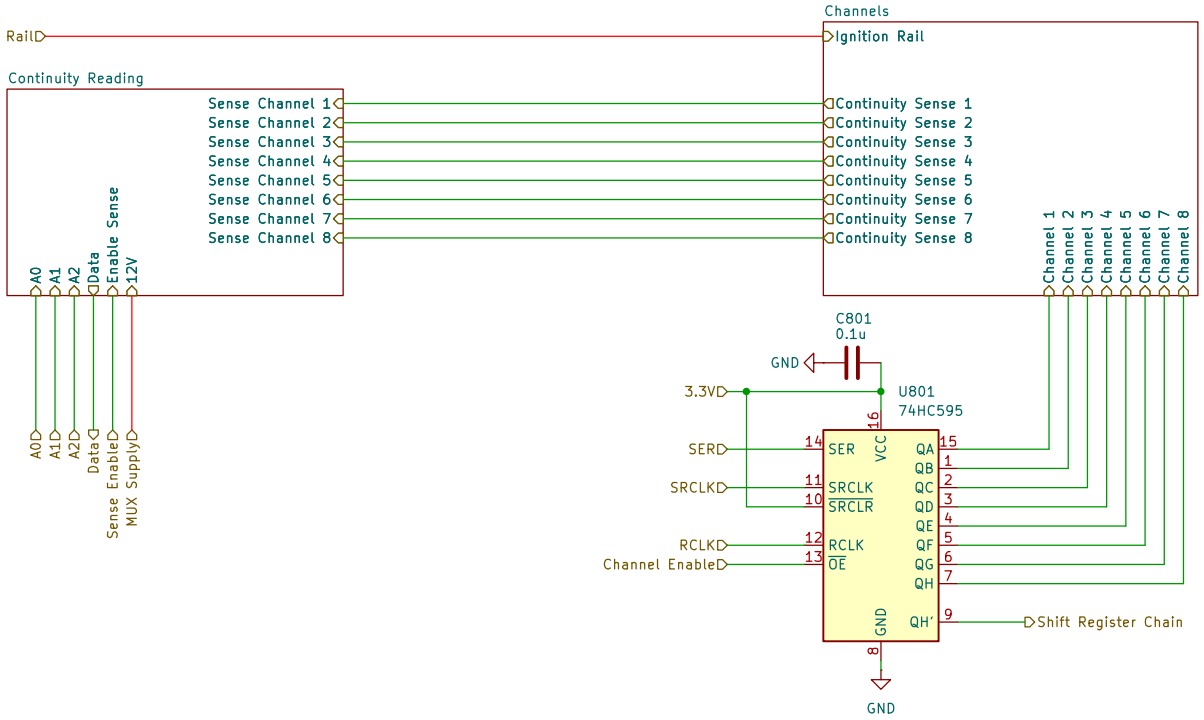
IGNITION CHANNELS 1–8 (low-side switches)
Channel n: J40n terminal (Pin 2 = Ignition Rail, Pin 1 = igniter return)
Q40n AO3400A N-FET pulls the igniter to GND when 'Channel n' is high
R40n 10k gate pull-down keeps the channel OFF while the driver is undriven
Drain node = 'Continuity Sense n' -> divider + mux (Continuity Reading)
An igniter only fires if the Ignition Rail is ON (Q301) AND Q40n is ON

Current measurement line – Continuity Sense
Channel On/Off control by Mosfets
One ignition current rail

Sheet: /ChannelBankA/Channels/
File: channels.kicad_sch

Title: Channels

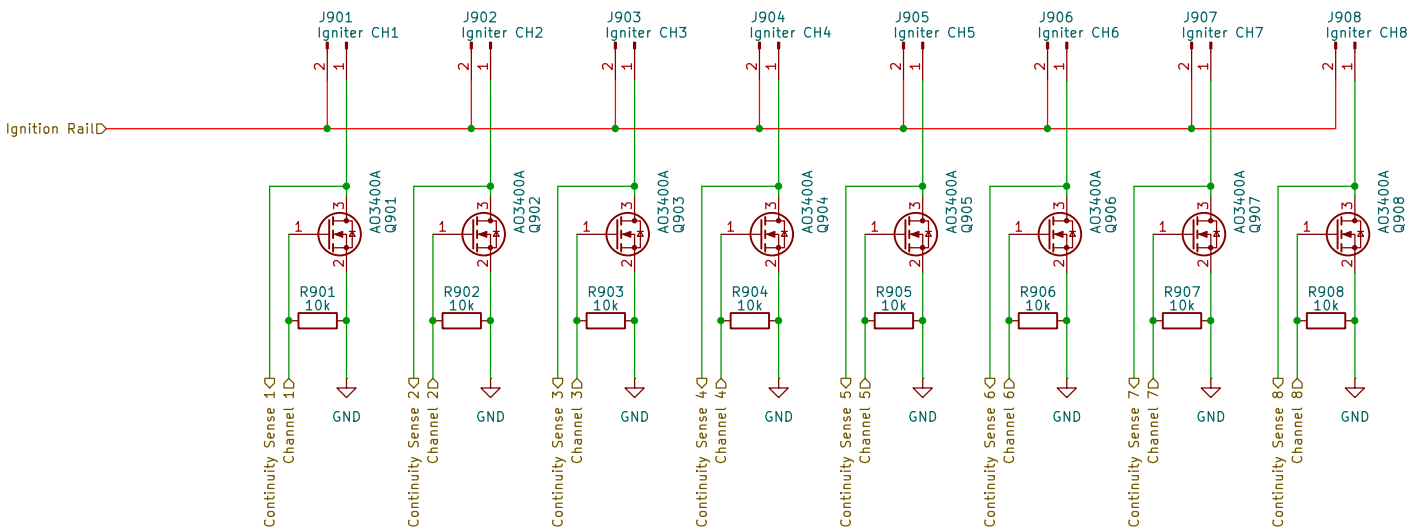
Size: A4	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/10



Sheet: /ChannelBankB/
File: channelBank.kicad_sch

Title: Channel Bank

Size: A4	Date: 2026-11-01	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/10



IGNITION CHANNELS 1–8 (low-side switches)
Channel n: J40n terminal (Pin 2 = Ignition Rail, Pin 1 = igniter return)
Q40n AO3400A N-FET pulls the igniter to GND when 'Channel n' is high
R40n 10k gate pull-down keeps the channel OFF while the driver is undriven
Drain node = 'Continuity Sense n' -> divider + mux (Continuity Reading)
An igniter only fires if the Ignition Rail is ON (Q301) AND Q40n is ON

Current measurement line – Continuity Sense
Channel On/Off control by Mosfets
One ignition current rail

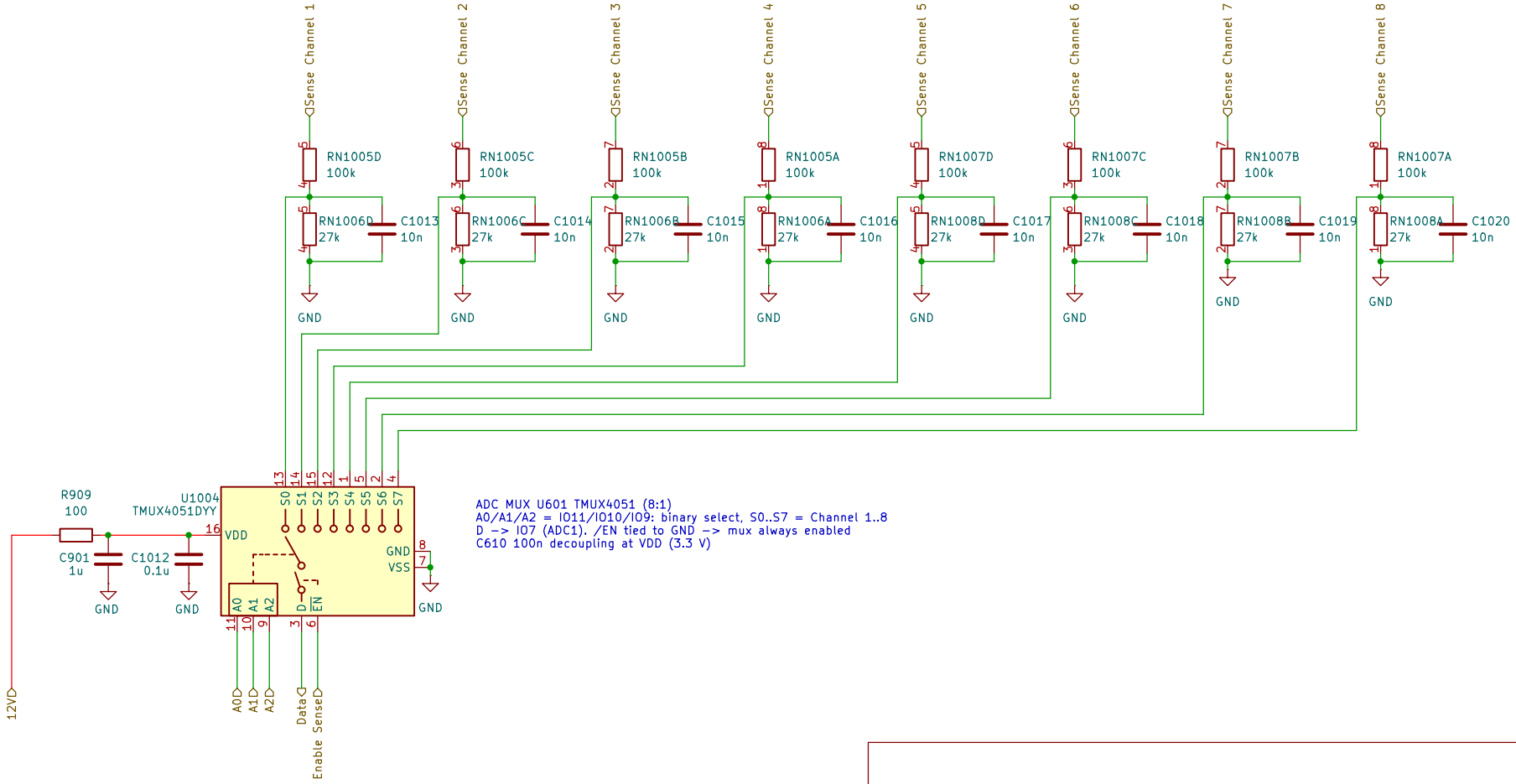
Sheet: /ChannelBankB/Channels/
File: channels.kicad_sch

Title: Channels

Size: A4	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/10

CHANNEL SENSE (one per channel)
Sense Channel n –> 100k / 27k divider + 10n –> mux input S(n–1)
Ch1–4: RN601 (100k) / RN602 (27k), C601–C604
Ch5–8: RN603 (100k) / RN604 (27k), C605–C608
Igniter connected –> node follows the rail voltage, open –> ~0 V

RAIL SENSE: Ignition Rail –> R601 100k / R602 27k, C609 10n
–> Sense Rail OUT –> IO8 (ADC). Same scaling as the channel inputs



ADC MUX U601 TMUX4051 (8:1)
A0/A1/A2 = IO11/IO10/IO9: binary select, S0..S7 = Channel 1..8
D –> IO7 (ADC1). /EN tied to GND –> mux always enabled
C610 100n decoupling at VDD (3.3 V)

TMUX4051 8:1 mux, A0-A2 select, D –> ADC, always enabled
8–CH VOLTAGE SENSE: 100k/27k divider (0-15.5 V –> 0-3.3 V), 10 nF filter (fc ≈ 750 Hz)

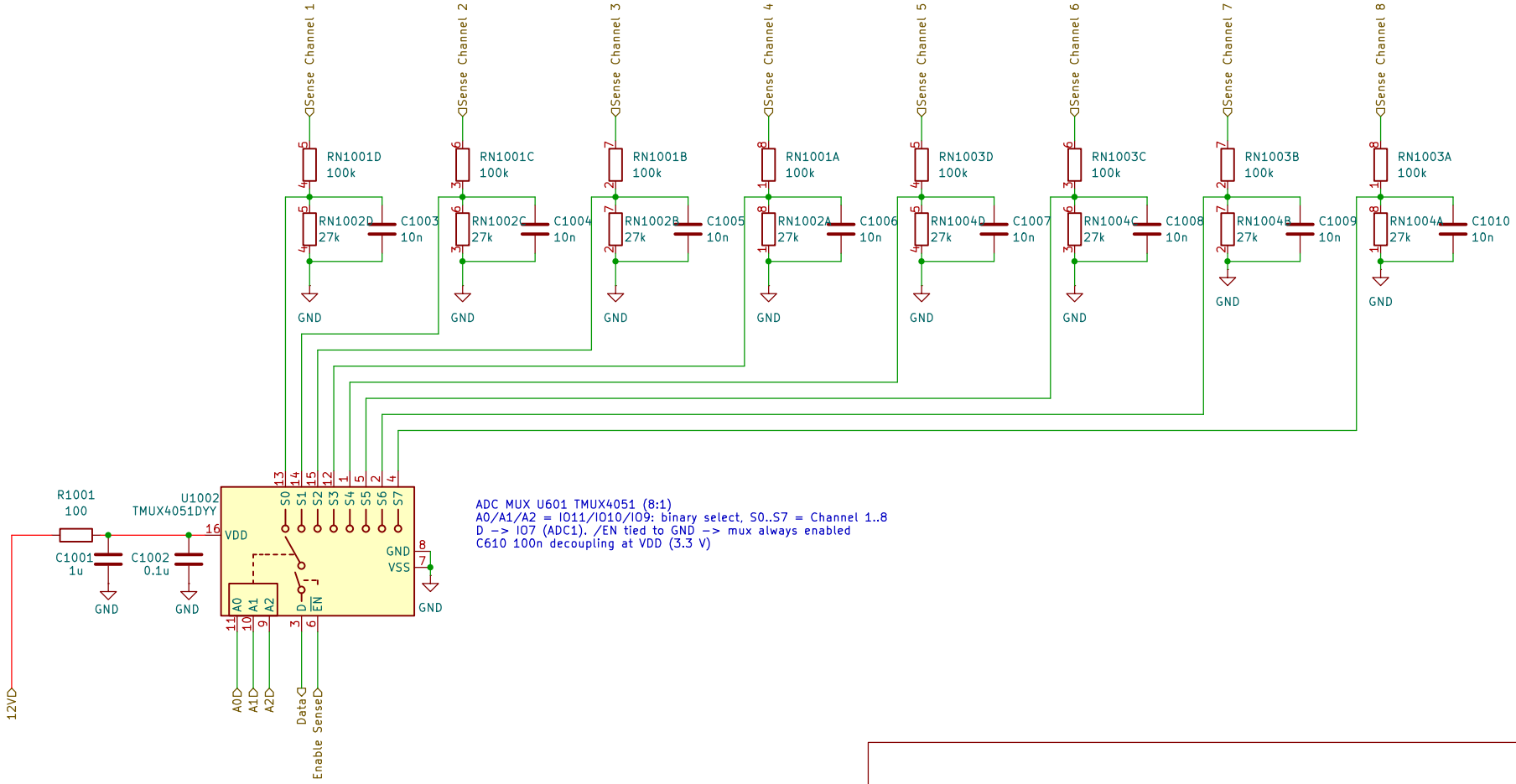
Sheet: /ChannelBankB/Continuity Reading/
File: ContinuityReading.kicad_sch

Title: Continuity Sensing

Size: A4	Date: 2026–10–26	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/10

CHANNEL SENSE (one per channel)
Sense Channel n –> 100k / 27k divider + 10n –> mux input S(n–1)
Ch1–4: RN601 (100k) / RN602 (27k), C601–C604
Ch5–8: RN603 (100k) / RN604 (27k), C605–C608
Igniter connected –> node follows the rail voltage, open –> ~0 V

RAIL SENSE: Ignition Rail –> R601 100k / R602 27k, C609 10n
–> Sense Rail OUT –> IO8 (ADC). Same scaling as the channel inputs



ADC MUX U601 TMUX4051 (8:1)
A0/A1/A2 = IO11/IO10/IO9: binary select, S0..S7 = Channel 1..8
D –> IO7 (ADC1). /EN tied to GND –> mux always enabled
C610 100n decoupling at VDD (3.3 V)

TMUX4051 8:1 mux, A0-A2 select, D –> ADC, always enabled
8–CH VOLTAGE SENSE: 100k/27k divider (0-15.5 V –> 0-3.3 V), 10 nF filter (fc ≈ 750 Hz)

Sheet: /ChannelBankA/Continuity Reading/
File: ContinuityReading.kicad_sch

Title: Continuity Sensing

Size: A4	Date: 2026–10–26	**Rev:**
KiCad E.D.A. 10.0.6		Id: 10/10